Project Phase: Scalability & Future Plan

Current System Limitations, Scalability & Roadmap

Date: 15 March 2026	**Team ID:** PNT2026TMID092418
Project Name: Global Olympic Sports Performance Analysis	**Maximum Marks:** 1 Mark

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	**Batch CSV Data Ingestion**	Requires manual batch triggers when importing newly concluded Olympic Games datasets.	**High**
2	**Descriptive Focus without Predictive Forecasting**	Dashboards analyze historical trends (1992–2026) but do not yet forecast 2028/2032 medal outcomes.	**Medium**
3	**Static Sports Investment Data Sources**	Financial investment figures are updated annually rather than via real-time financial APIs.	**Medium**

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	**User Load**	Supports concurrent internal analyst views via Tableau Desktop / Public.	Deploy to Snowflake & Tableau Server with auto-scaling virtual warehouses to support thousands of global concurrent users.
2	**Data Storage**	26.5 MB compressed historical dataset stored in Snowflake staging schemas.	Scale Snowflake storage tier dynamically to ingest multi-terabyte real-time biometric and granular event-level tracking data.
3	**Performance**	Sub-1.2 second dashboard filter queries.	Implement Snowflake Materialized Views and automated query acceleration services to maintain sub-500ms query performance.
4	**Security**	Standard user role access in Snowflake.	Implement Role-Based Access Control (RBAC), row-level security for sensitive NOC financial data, and SSL encryption in transit/at rest.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Machine Learning Predictive Models for 2028 LA & 2032 Brisbane Olympic Medal Projections	Q3 2026	Empowers NOC strategists to forecast medal yields based on athlete training trajectories and funding allocations.
Phase 3	Automated Real-Time Pipeline Integration with IOC & Financial APIs	Q4 2026	Eliminates manual CSV ingestion, providing real-time performance and budget tracking.
Phase 4	Mobile-Native Analytics App & Automated Executive PDF Report Generator	Q1 2027	Delivers instant dashboard access and automated policy memos to executive stakeholders on mobile devices.